RECOVERY LIVES IN U, CONDITIONING ACTS IN R: WHERE ARBITRARY-SCALE SUPER-RESOLUTION GAINS COME FROM

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ABSTRACT

Arbitrary-scale super-resolution (ASISR) networks pair an image encoder with an implicit decoder that also receives the target pixel size, the *cell*. Decoder and cell designs keep multiplying, yet their gains shrink at large scales. We explain this with two frequency coordinates. Per-band recovery is a function of frequency normalised by the low-resolution (LR) Nyquist limit, u. Four decoder designs on a shared encoder give the same recovery to within 0.02, on crops and on full benchmark images. Longer training, a stronger encoder (RDN, SwinIR) and a larger decoder MLP raise recovery only for $u \gtrsim 0.75$. The cell, by contrast, acts on the output as a low-pass filter in high-resolution frequency r. From the orthogonality principle we derive that its cost is the recovered energy it filters away. A fixed r moves to larger u as the scale grows, where little has been recovered, so the cell's leverage vanishes. Doubling the cell costs 2.3 dB at $\times 2$, 0.13 dB at $\times 4$ and nothing measurable at $\times 6$. Removing the cell input from training costs 0.10 dB at $\times 2$ and is indistinguishable from zero from $\times 3$ on. All predictions were registered before the experiments, and we report those that failed.

***Index Terms*—** Arbitrary-scale super-resolution, implicit neural representation, frequency analysis, scale conditioning

1. INTRODUCTION

LIIF [1] reconstructs an image at continuous coordinates by decoding local encoder features together with the query offset and the target cell size. Many decoders have followed: local Fourier estimators [2], integrated positional encoding [3], implicit attention [4], neural operators [5], Gaussian splatting [6] and filtered heat fields [7]. Their gains are small. A recent benchmark reports saturation, with improvements that follow model size, compute and data rather than decoder form [8]. A spectral decoder reports leads over strong baselines that vanish at large scales, without an explanation [9]. Frequency analyses of SR networks characterise low-pass behaviour [10] and sinusoidal responses [11], and study how frequency representations transfer across resolutions [12]. None of them locates *where* in frequency a decoder or the cell can still change the output, or why that room shrinks with scale.

We answer with two coordinates. For scale s and HR frequency r (cycles per HR pixel), $u = 2sr$ is the frequency relative to the LR Nyquist limit. We make three claims.

1. **Recovery lives in u.** Per-band recovery $R(u)$ does not depend on decoder design. What moves it is learned capacity (training time, encoder or decoder MLP), and only above $u \approx 0.75$.
2. **Conditioning acts in r.** Changing the cell filters the native output by an approximately scale-independent low-pass $w(r)$. Its cost equals the recovered energy at the filtered frequencies (Eq. (3)), so the cell's leverage decays with scale.
3. **Consequences.** At $\times 3$ and above, the cell input can be removed at no measurable cost. Decoder form is not a lever, whereas capacity is, wherever it sits.

That the cell input can sometimes hurt is known [3]. Our contribution is a quantitative account of where decoder- and cell-side changes can act at all. Each experiment fixed its code, predictions and scoring before running; Table 2 lists every outcome.

2. SETUP

Models. LIIF, LTE [2] and SC-INR (an LTE variant whose cell enters through an analytic sinc weight) share an EDSR-baseline encoder [13] and the LIIF recipe: DIV2K [14], scales drawn from $\mathcal{U}(1, 4)$, 1000 epochs, three seeds. GaussianSR [6] adds a fourth decoder on the same encoder. For matched comparisons we retrain LIIF for 200 epochs with one recipe, changing only one component: an RDN [15] or SwinIR [16] encoder, a smaller decoder MLP (2×64 instead of 4×256), or no cell input.

Data. We use DIV2K validation images 0869–0900, not used for any model selection, and full Set14 [17], BSD100 [18] and Urban100 [19]. LR inputs come from anti-aliased bicubic downsampling at $\times\{2, 3, 4, 6\}$; $\times 6$ lies outside the training range. Unless stated otherwise we evaluate centre 240^2 crops.

Measures. After a 4-pixel shave, the error and the target spectrum are split by Chebyshev frequency radius into u-bins of width 0.25. We define $R(u) = 1 - \mathrm{MSE}_u / E_u$, where E_u is the target energy in the bin. To probe the cell, we multiply the native cell by $m \in [\frac{1}{2}, 2]$ at fixed input and query.

3. RECOVERY LIVES IN U

Below the LR Nyquist limit, networks and bicubic interpolation both recover nearly everything (Fig. 1a). The networks' advantage lies in $0.75 \lesssim u \lesssim 2$: at $u = 1$, $R = 0.68$ against 0.20 for bicubic; at $u = 1.5$, 0.36 against 0.04. Curves for different scales nearly coincide. This is partly generic: bicubic interpolation collapses in u too, reflecting roughly scale-invariant image statistics and a degradation defined on the LR grid. The collapse is also only approximate (Table 2, first row). Our claims therefore rest on which components can *move* $R(u)$.

Decoder design does not. On a shared encoder, LIIF, LTE, SC-INR and GaussianSR differ by at most 0.010 in every bin (95% image-bootstrap interval $[0.007, 0.016]$; Fig. 1b). On full Set14, BSD100 and Urban100 images the spread stays below 0.019 in all 54 dataset–scale–bin cells. Accordingly, their PSNR differs by only 0.03–0.06 dB.

Learned capacity does, above $u \approx 0.75$. Three interventions change $R(u)$, and all three act only there (Fig. 1c). From epoch 100 to 1000, R rises by up to 0.045 for $u \geq 1$ while changing by at most 0.008 below $u < 0.75$. With the recipe fixed, an RDN encoder raises R by up to about 0.05 for $u \geq 1$ in both seeds, and SwinIR does likewise, with $u < 0.75$ unchanged. Shrinking the decoder MLP lowers R over the same range. Table 1 gives the PSNR effects.

4. CONDITIONING ACTS IN R

Let P_1 be the native output and P_m the output with the cell multiplied by m, both projected onto band b. Empirically,

$$P_{m,b} \approx w_m(r_b)\, P_{1,b}, \qquad w_m(r) \in \mathbb{R}, \tag{1}$$

where w_2 is a low-pass in HR frequency with its minimum at $r \approx 0.3$ for all three LIIF-family decoders (Fig. 2a). With $e_b = T_b - P_{1,b}$ the native error, Eq. (1) gives the exact expansion

$$\Delta\mathrm{MSE}_b = 2(1-w)\,\mathrm{Re}\langle e_b, P_{1,b}\rangle + (1-w)^2\|P_{1,b}\|^2. \tag{2}$$

For an output equal to the conditional mean, the orthogonality principle gives $\langle e_b, P_{1,b}\rangle = 0$. The first-order term then vanishes, which is what a calibrated native cell means. The emitted energy also equals the recovered energy, $\|P_{1,b}\|^2 = R_b E_b$. Hence

$$\Delta\mathrm{MSE}(s) \approx \sum_r \big(1 - w(r)\big)^2\, R(2sr)\, E(r). \tag{3}$$

The two coordinates meet here: the cell supplies a filter in r, the network supplies recovery in $u = 2sr$. Because R falls with u, the same filter removes less and less recovered energy as s grows.

Checking the derivation. The networks are trained with an L_1 loss, so orthogonality can only hold approximately. On the DIV2K images we find $|\langle e_b, P_{1,b}\rangle|/\|P_{1,b}\|^2 \leq 0.15$ in all 105 bins with $u < 2$, for all three decoders. For $m = 2$, Eq. (3) predicts the measured cost at $\times 3$ and $\times 4$ to within a factor of 0.71–0.84. It underpredicts consistently, because the cell also changes the output beyond a per-band scalar. At $\times 2$ the shortfall reaches about $2\times$ on Set14. For small mismatch ($m = 4/3$) the first-order term dominates and the approximation fails.

Table 1. Matched LIIF training (200 epochs, mean of two seeds): PSNR change (dB) relative to the EDSR-baseline LIIF. All 95% image-bootstrap intervals exclude zero except those marked $^\circ$. †One seed, equal 50 epochs.

Change	$\times 2$	$\times 3$	$\times 4$	$\times 6$
decoder MLP 2×64	-0.21	-0.17	-0.13	-0.11
no cell input	-0.10	-0.03°	-0.02°	-0.00°
RDN encoder	$+0.30$	$+0.23$	$+0.23$	$+0.18$
SwinIR encoder†	$+0.17$	$+0.18$	$+0.12$	$+0.07$

Consequences. The cost of a doubled cell peaks at $r \approx 0.25$ for every LIIF-family model, seed and scale (Fig. 2b), and it tracks recovered energy (Spearman 0.77–0.78). It decays monotonically with scale: 2.3 dB at $\times 2$, 0.40 dB at $\times 3$, 0.13 dB at $\times 4$ and 0.01 dB at $\times 6$ (interval includes zero; Fig. 2c). The decay also holds for GaussianSR and for an external RDN-BTC model [20], although their cost peaks at other r. The native cell is close to optimal: no fixed multiplier improves on it by more than 0.017 dB.

5. WHAT IS A LEVER

Table 1 tests the consequences by retraining. Removing the cell input costs 0.10 dB at $\times 2$, where the doubled-cell probe showed large leverage. From $\times 3$ on its effect is indistinguishable from zero, as Eq. (3) predicts. Stronger encoders help at all scales. Before training we had also predicted that a much smaller decoder MLP would be harmless at large scales. That failed: it costs 0.11–0.21 dB at every scale in both seeds, again only for $u \gtrsim 0.75$ (Fig. 1c). The decoder MLP thus stores part of the above-Nyquist prior. The accurate reading is that decoder *form* is not a lever, since four designs give identical $R(u)$, while learned capacity is, wherever it sits.

6. LIMITATIONS AND CONCLUSION

We study bicubic degradation, LIIF-style training, and convolutional and Transformer encoders. The matched arms use two seeds and short schedules. The filter model holds for large mismatch only, and the frequency at which the cell acts depends on the decoder family. Within these limits, ASISR gains come from what the network learns above the LR Nyquist limit. Learned capacity, not decoder form, sets that prior. The cell is an HR-frequency control whose reach is the recovered energy it filters. Eq. (3) thus tells in advance at which scales

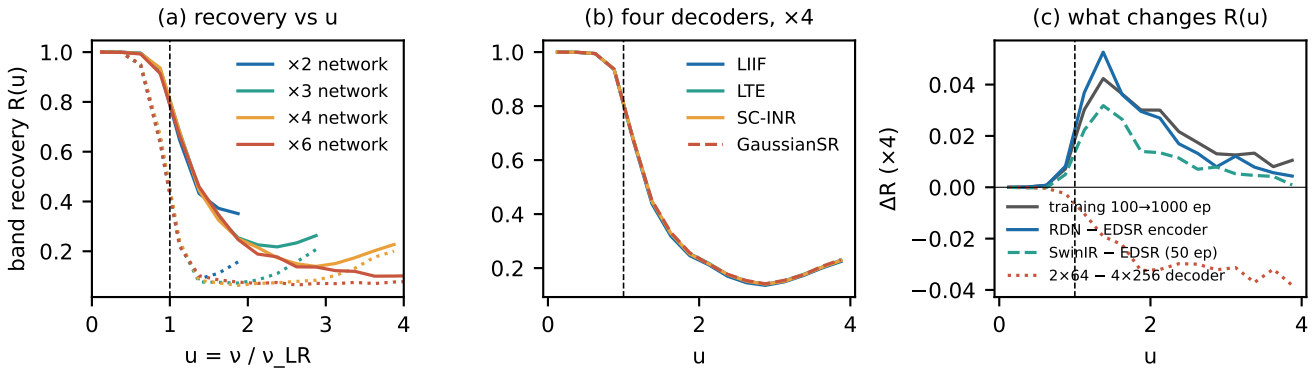


Fig. 1. Recovery in the LR-normalised coordinate u. (a) $R(u)$ at four scales (solid: mean of three decoders and three seeds; dotted: bicubic). (b) Four decoders on the same encoder at $\times 4$. (c) Change in $R(u)$ at $\times 4$ from longer training, stronger encoders and a smaller decoder MLP.

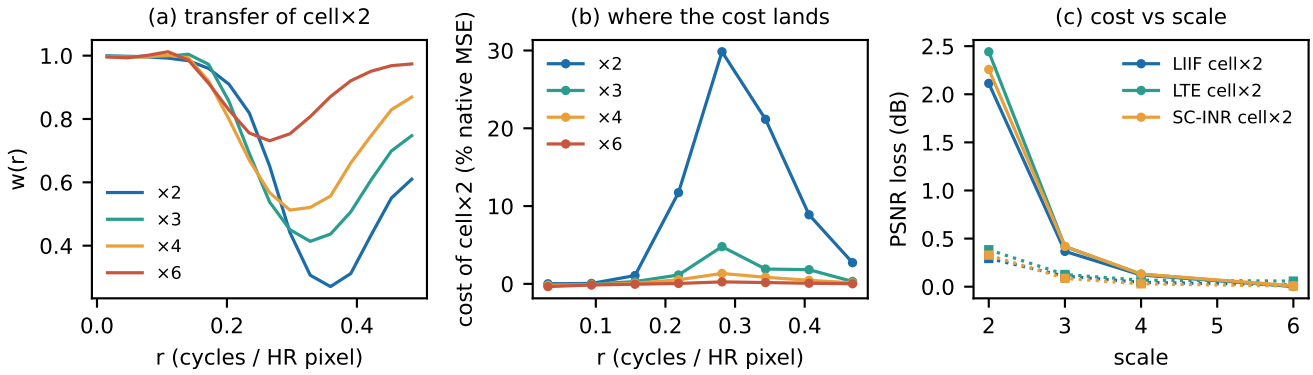


Fig. 2. Conditioning in the HR coordinate r. (a) Spectral transfer $w_2(r)$ of doubling the cell (mean of three decoders). (b) Where its cost lands (LTE). (c) PSNR cost of doubling (solid) or halving (dotted) the cell, by scale.

a new cell or footprint design can matter: only where $R(2sr)$ remains substantial at the frequencies the design alters.

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Table 2. Registered predictions and outcomes. Fractions count bins, models, seeds or scales as specified before each run.

Prediction	Outcome
$R(u)$ spread across scales ≤ 0.05 (crops; full images)	45/54; 53/72 **fail**
four decoders within 0.03 (crops; full images)	36/36; 54/54
training raises R only for $u \geq 1$	12/12; 9/9
RDN / SwinIR raise R only for $u \geq 1$	pass / pass
seed-2 replication of matched arms	11/12; pass
cell$\times 2$ cost peaks at $r \in [0.19, 0.375)$	36/36
cell cost tracks recovered energy ($\rho \geq 0.7$)	0.77–0.78
cell leverage decays with scale	9/9
Eq. (3) within $\times[0.7, 1.3]$ at $\times 3/\times 4$	6/6
near-orthogonality (≤ 0.15)	105/105
native cell is the optimum	20/27 **fail**
a too-small cell costs more than a too-large one	33/144 **fail**
$\geq 80\%$ of cell cost at $u \geq 1$	75% **fail**
small decoder harmless at $\times 4$+	**fail**
no-cell cost ≤ 0.03 dB at $\times 4$+ (seed 1)	**fail** (narrow)
encoder gain exceeds small-decoder loss	pass

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